

Unlocking Behavioral Intelligence in Airline Loyalty Programs

Customer Churn Prediction, Segmentation & Smart Retention — Technical Report

Prepared by Tanishq Gupta | AI/ML Portfolio Project | Dataset: Airline Loyalty Program (Customer Demographics, Loyalty & Monthly Flight Activity)

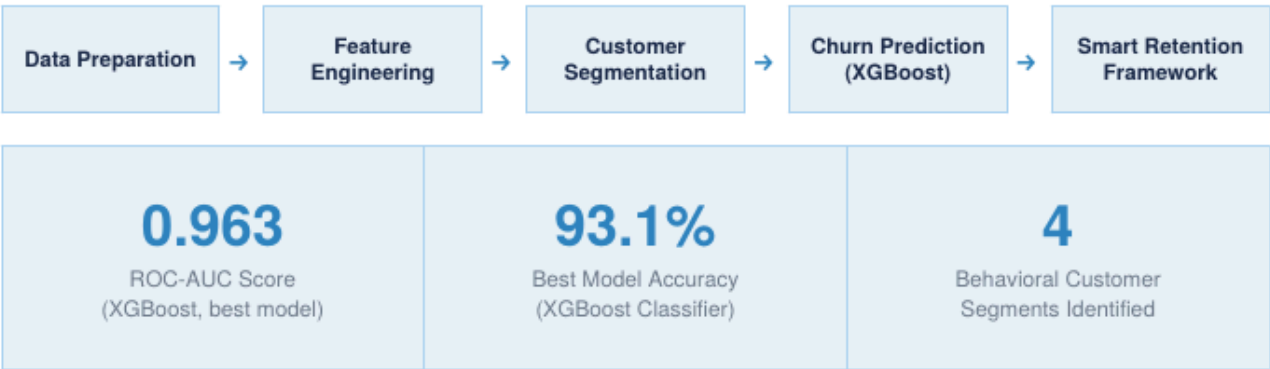
1. Executive Summary

Customer retention is one of the most important drivers of profitability in the airline industry. Acquiring new customers is significantly more expensive than retaining existing ones, making early identification of disengaged customers a strategic priority. This project develops an end-to-end analytical framework that combines customer demographics, loyalty information, and monthly activity records into behavioral profiles, then applies machine learning to translate those profiles into actionable retention decisions.

Project Objectives

- **Predict customer churn** using behavioral and demographic signals.
- **Identify meaningful customer segments** based on natural behavioral groupings.
- **Recommend targeted retention strategies** tailored to each segment.

Analytical Pipeline



Key Finding

Customer behavior is a far stronger indicator of future churn than demographic characteristics or even Customer Lifetime Value (CLV). Booking activity, recency, and behavioral trend variables consistently outperformed static attributes such as salary, education, or loyalty card tier across every stage of the analysis.

2. Problem Formulation

Acquiring new customers is significantly more expensive than retaining existing ones, making proactive churn management essential. The project was framed around three core business questions, each translating an analytical output into an actionable business decision.

Business Question	Description
Churn Prediction	Which customers are likely to stop engaging with the airline?
Customer Segmentation	Are there naturally occurring groups of customers with different behavioral patterns?
Smart Retention	How can marketing resources be allocated effectively to reduce customer attrition?

3. Data Description & Preparation

3.1 Data Sources

Dataset	Contents
Customer Loyalty History	Demographic information, loyalty tier, Customer Lifetime Value, and cancellation records.
Customer Flight Activity	Monthly records of flights booked, distance travelled, and points accumulated or redeemed.
Calendar Dataset	Mapping of months to seasons and quarters for seasonality analysis.
Data Dictionary	Definitions and descriptions of all variables used across datasets.

3.2 Data Cleaning & Preparation

- **Missing values** representing absence of activity were replaced with zero.
- **Infinite values** generated by ratio-based features were removed or replaced appropriately.
- **Aggregation:** monthly activity records were aggregated to customer level to obtain a single profile per customer.
- **Avoiding data leakage:** only information available at the time of prediction was used to construct features, ensuring model performance reflects realistic business scenarios.

4. Feature Engineering

Rather than relying only on demographics, the analysis emphasized customer behavior through value, recency, trend, and seasonality variables.

4.1 Customer Value Variables

Customer Lifetime Value (CLV)	Total Bookings
Average Distance Travelled	Total Points Accumulated
Total Points Redeemed	—

4.2 Recency

Recency was defined as the number of months between the customer's last flight and the end of 2018. Customers with larger recency values were considered less engaged.

4.3 Trend Variables

Behavioral changes were captured through three trend variables that provide insight into customer momentum rather than relying only on absolute values.

Variable	What It Captures
Booking Trend	Whether recent booking activity has increased or decreased relative to historical activity.
Accumulation Trend	Changes in point accumulation behavior over time.
Redemption Trend	Changes in point redemption behavior over time.

4.4 Seasonality Index

Seasonal booking patterns were analyzed across Spring, Summer, Autumn, and Winter. This variable helps distinguish consistent travelers from customers whose activity is concentrated in particular seasons.

5. Exploratory Data Analysis & Customer Value

Correlation analysis revealed that demographic variables such as salary exhibited very weak relationships with churn and Customer Lifetime Value, while behavioral variables showed considerably stronger associations with customer engagement.

Pareto Analysis	Customers were ranked by Customer Lifetime Value to evaluate the concentration of value across the customer base. The analysis confirmed the Pareto principle: a relatively small group of customers contributes a disproportionately large share of total value.
RFM Analysis	Customers were evaluated using Recency, Frequency, and Monetary value. RFM analysis helped identify VIP customers and provided additional insight into customer value distribution.

6. Customer Segmentation

K-Means clustering was used to group customers according to behavioral characteristics. The elbow method and silhouette scores were used to evaluate different values of K; a four-cluster solution was selected because it provided meaningful, interpretable customer groups while avoiding excessive complexity.

Segment	Behavioral Profile
Champions / Premium Travelers	Highest booking frequency, long travel distances, strong positive booking trends, and high seasonality.
Active Regular Customers	Stable engagement, consistent travel behavior, and moderate activity levels.
Occasional Flyers	Low booking frequency, short travel distances, and moderate engagement.
Dormant Customers	High recency values, declining activity, and negative behavioral trends.

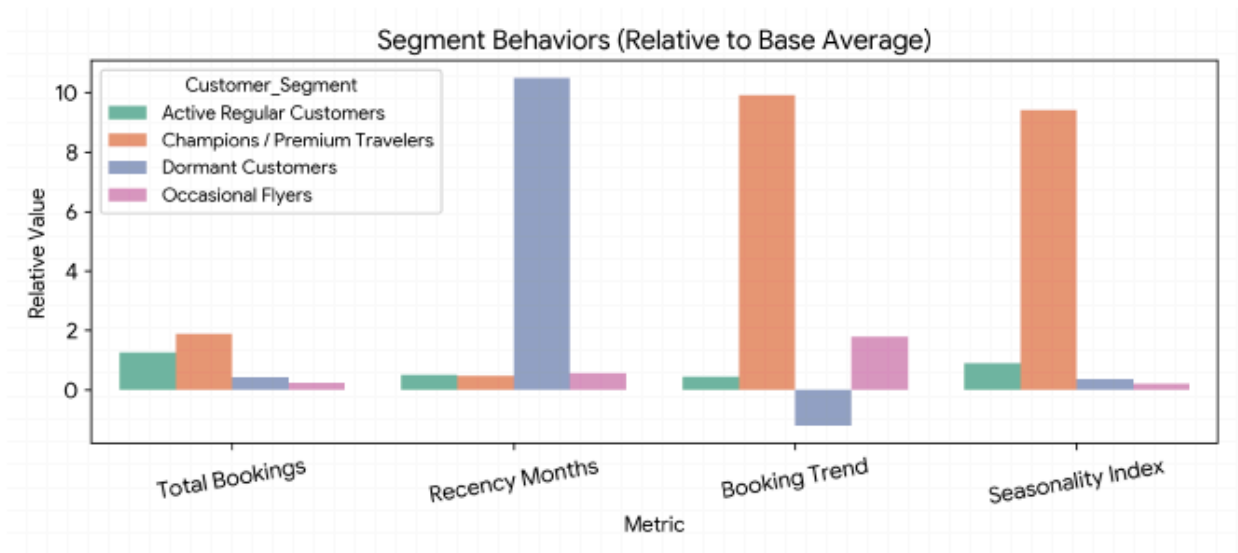


Figure 1. Segment behavior profiles relative to the base average across key metrics.

Demographic Insights: cross-tabulation analysis revealed minimal differences across gender, education, marital status, and loyalty card type. This suggests that behavior, rather than demographics, drives customer segmentation.

7. Churn Prediction Model

Three machine learning models were evaluated for churn classification:

Model	Accuracy	Recall	F1 Score	ROC-AUC
Logistic Regression	91.6%	—	—	—

Random Forest	92.8%	—	—	—
XGBoost (selected)	93.1%	81.6%	0.746	0.963

XGBoost achieved the strongest overall performance and was selected as the final model for the prototype dashboard.

8. Key Drivers of Churn

Feature importance analysis showed that customer behavior dominates demographic information in predicting churn.

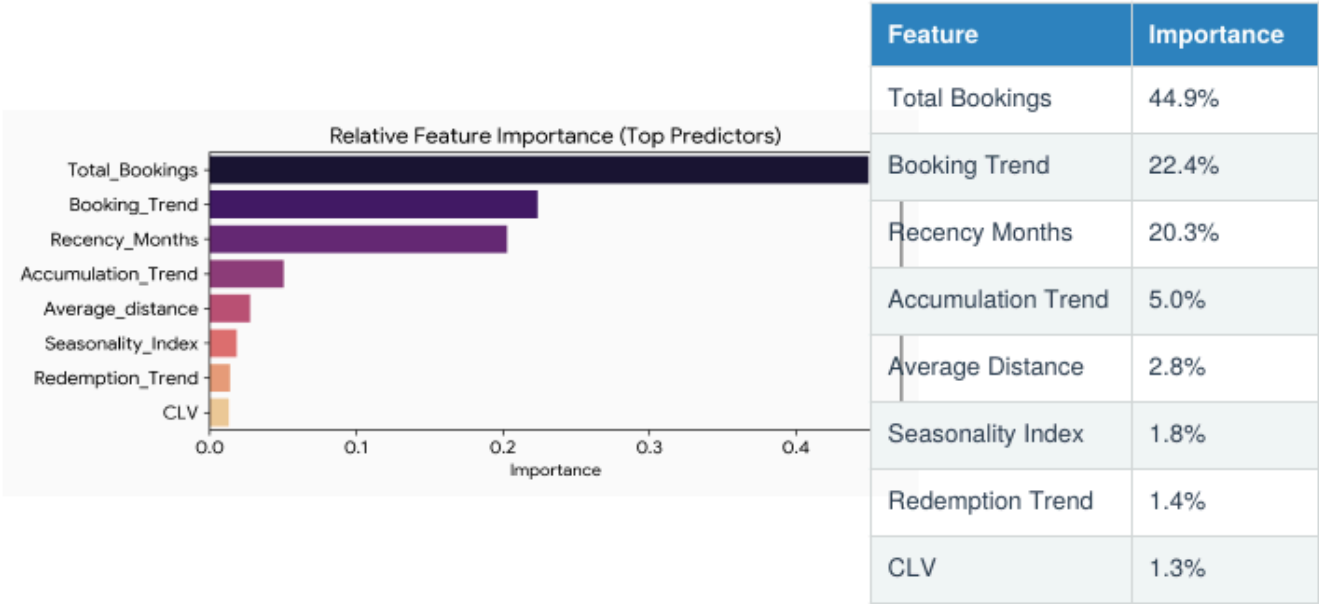


Figure 2. Relative feature importance for the top predictors of churn (XGBoost).

Interesting Findings

Customer Lifetime Value had surprisingly low predictive importance, suggesting that declining behavior provides stronger signals of future churn than historical customer value. Demographic variables contributed very little to customer differentiation overall.

9. Smart Retention Framework

Segment	Recommended Retention Strategies
Champions / Premium Travelers	• Priority services • Exclusive promotions • Tier upgrades • Premium reward programs

Active Regular Customers	• Maintain engagement • Personalized travel offers • Seasonal campaigns
Occasional Flyers	• Bonus miles • Promotional discounts • Targeted destination recommendations
Dormant Customers	• Reactivation campaigns • Personalized emails • Limited-time incentives • Win-back promotions

10. Prototype Dashboard

A Streamlit dashboard was developed to provide a user-friendly interface for non-technical users, covering customer segmentation, churn prediction, smart retention recommendations, feature importance analysis, and business insights.

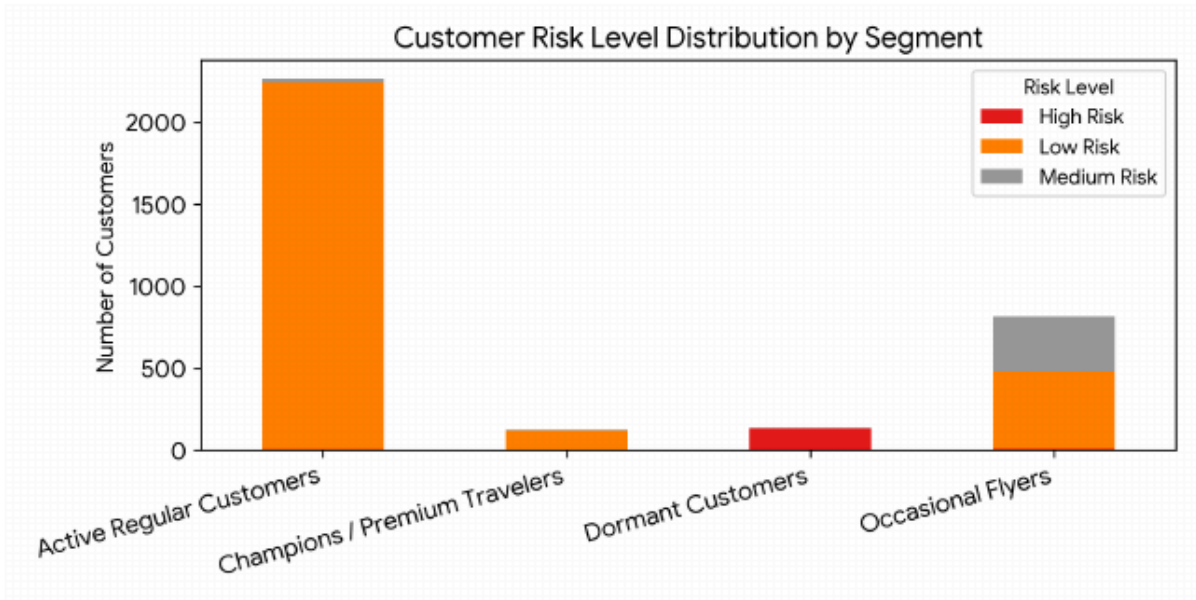


Figure 3. Distribution of customer risk levels by behavioral segment, as surfaced in the dashboard.

11. Business Recommendations

1. Prioritize Behavioral Monitoring	Customer behavior is a stronger predictor of churn than demographic variables. Organizations should continuously monitor booking trends and recency indicators.
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2. Protect High-Value Customers	Champions and Premium Travelers generate disproportionate value. Retention efforts should focus on preserving these relationships through exclusive rewards and personalized experiences.
3. Intervene Early	Customers exhibiting declining booking trends should receive targeted incentives before complete disengagement occurs. Proactive intervention is significantly more cost-effective than reacquiring lost customers.

12. Conclusion

This project demonstrates that customer behavior contains valuable signals that can be leveraged to predict churn and understand customer value. By combining segmentation with machine learning, the airline can move from reactive customer management to proactive retention strategies. The resulting framework provides both predictive capability and actionable business insights, enabling the organization to strengthen customer relationships and maximize the long-term value of its loyalty program.